

Beta Decay Spectra of Sr-90 and Y-90: Characterizing a Detector Response Model

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1. Introduction

1.1 Project Objective and Question

Beta decay theory provides a thoroughly defined prediction for the energy distribution of electrons emitted by radioactive nuclei. However, radiation detectors do not measure all emitted particle energies directly. Instead, they record deposited energy, a quantity that can differ substantially from the true emission energy due to things like electron escape and backscattering occurring in the detector. As a consequence, the observed spectrum may be dramatically reshaped with relation to the actual underlying physics.

This project investigates the beta decay spectrum of a combined Sr-90 radioactive source measured using a CsI scintillation detector. The central research question addressed is

How does the detector response of a CsI scintillation detector modify the observed beta decay spectrum of an Sr-90 source, and can a detector response model quantitatively reproduce the measured spectrum from first-principles theory alone?

To address this question, the following steps are taken:

- Construct theoretical beta decay spectra for Sr-90/Y-90 using a relativistic equation that incorporates an exact Fermi function

- Combine the individual isotopes spectra under equilibrium to obtain the expected total emission spectrum.
- Compare the theoretical spectrum to the experimentally measured spectrum, and identify any natures of residuals that could be detector induced.
- Begin to develop a semi-empirical detector response model that incorporates energy resolution, energy-loss tails, and detection efficiency effects
- Fit the model's free parameters to the experimental spectrum using nonlinear least squares regression and evaluate the quality of the resulting fit, along with its real meaning.

Through this, the project will aim to distinguish between observed effects that are intrinsic nuclear physics and ones that align more with detector behavior distortions, and eventually demonstrating that the observed spectrum shape past the peak and tail is governed primarily by detector response rather than the underlying beta decay processes.

1.2 Physics of Beta Decay

Beta decay is a weak nuclear interaction in which a neutron in an atomic nucleus 'decays' is converted into a proton and electron (beta particle), and an antineutrino:



Because beta decay is a three-body process, the emitted electron exhibits a continuous energy spectrum that extends from zero up to a maximum endpoint energy E_0 . This is in contrast to decay alpha or gamma emission, where discrete energy packets are emitted.

The shape of the beta energy spectrum is derived from Fermi's golden rule. Accounting for the relativistic effects, the differential decay rate per unit energy is proportional to:

$$dN/dE \propto F(Z, E) \cdot p \cdot E_{\text{tot}} \cdot (E_0 - E)^2$$

Here, E is the electron kinetic energy, $p = (1/c) \sqrt{(E_{\text{tot}}^2 - (m_e c^2)^2)}$ is the relativistic electron momentum, and $E_{\text{tot}} = E + m_e c^2$ is the total electron energy. The factor $p \cdot E_{\text{tot}}$ represents the electron phase space, while $(E_0 - E)^2$ represents the antineutrino phase space. As the electron energy approaches the endpoint E_0 , the antineutrino energy approaches zero, leaving very few available final states for the antineutrino. By Fermi's golden rule, the transition rate vanishes, creating what we observe as the endpoint cutoff of the beta decay spectrum.

The emitted electron also experiences an electromagnetic Coulomb interaction with its causal nucleus. For β^- decay, this attractive potential promotes a lower-energy electron emission, shifting the spectrum toward those energies. This Coulomb correction is encoded in the Fermi function $F(Z, E)$. Since both Sr-90 and Y-90 have endpoint energies that exceed the electron rest mass energy we derive from $E = mc^2$, a fully relativistic calculation is implemented, obtained by solving the Dirac equation in a Coulomb potential with finite nuclear size. The exact Fermi function used in this work is:

$$F(Z, E) = [2(1+S) / \Gamma(1+2S)^2] \cdot (2p\rho)^{(2S-2)} \cdot e^{(\pi\eta)} \cdot |\Gamma(S+i\eta)|^2$$

where $S = \sqrt{1-(\alpha Z)^2}$, $\eta = \alpha Z E_{\text{tot}} / p$, $p = R p / \hbar c$, α is the fine structure constant, Z is the proton number of the daughter nucleus, and $R \approx 6/5 A^{(1/3)}$ is the nuclear radius. The full theoretical spectrum is then:

$$dN/dE \propto F(Z, E) \cdot p \cdot E_{\text{tot}} \cdot (E_0 - E)^2$$

1.3 Sr-90 and Y-90 Decay Chain and Secular Equilibrium

The radioactive source used in this experiment is initially composed of pure Sr-90. Strontium-90 undergoes β^- decay to Yttrium-90 (Y-90), which itself undergoes β^- decay to stable Zirconium-90 (Zr-90). The decay chain is:



The endpoint energies for the two decays are $E_0(\text{Sr-90}) = 546 \text{ keV}$ and $E_0(\text{Y-90}) = 2280 \text{ keV}$. Because the half-life of Sr-90 is 28.91 years and the half-life of Y-90 is only 64.05 hours, the two isotopes can be assumed to be in secular equilibrium. In this equilibrium, the activity of the daughter (Y-90) equals that of the parent (Sr-90), and both contribute equally to the measured spectrum:

$$N_{\text{true}}(E) = N_{\text{Sr}}(E) + N_{\text{Y}}(E)$$

Because Sr-90 only emits electrons with energies up to 546 keV, and because Y-90 spans the full range from 0 to 2280 keV, the combined theoretical spectrum exhibits elevated counts below 546 keV (where both isotopes contribute) and Y-90 dominance above 546 keV. The theoretical combined spectrum forms the baseline expectation for which experimental data is compared, before any detector effects are considered

Figure 1 (Data and Analysis section, pg 6) shows the normalized individual and combined theoretical spectra, confirming the expected behavior. The combined spectrum inherits a double-peaked structure at low energies and the long high-energy tail of Y-90.

1.4 Motivation

The experiment serves to highlight a fundamental principle of experimental nuclear and particle physics, where the response of any detector can mediate what is actually observed. Even a well-understood nuclear process like beta decay can appear dramatically different depending on the detector used to measure it. By constructing a detector response model and fitting it to data, this project brings insight regarding the detector itself and demonstrates the methodology by which physicists can “deconvolve” detector effects from their measurements.

2. Procedure

2.1 Detector and Source

A CsI crystal scintillation detector was used to record the beta emission spectrum from the Sr-90 source. In a CsI scintillator, incident charged particles deposit energy by exciting the crystal lattice, which emits photons. These photons are detected by a photodetector, and the resulting electronic signal is proportional to the energy deposited in the crystal. The detector outputs count data across a discrete count of energy bins.

The source had an initial activity of 0.5 μ i as of March 1, 2013, and was sealed in a small plastic container. To maximize the number of detected beta particles, the plastic cover of the source container was opened and the detector was placed directly on top of the source. This minimized the air gap between source and detector face and maximized geometric solid-angle coverage.

2.2 Data Collection

Data was collected across a total of five experimental runs in the same session. The detector recorded energy-binned count data for ~900 seconds per run. The first four runs (spectra 0 through 3) were performed with the Sr-90/Y-90 source in place. For the fifth run, the source was removed from the detector vicinity, and 900 seconds of background radiation data were recorded without the source present.

An anomaly was observed during Spectrum 2, where the count magnitude was lower than in the other three source spectra. This was attributed to the detector physically slipping off the top of the source container for a brief interval during that trial, reducing the effective collection solid angle and thus the detected count rate. Despite this, all four source spectra were retained and averaged to improve overall counting statistics.

It should be noted that only a single background spectrum was collected rather than averaging over multiple background trials. This introduces a larger uncertainty in the background subtraction than would otherwise be present. Future work should collect multiple background trials and average them in the same manner as the source spectra.

2.3 Data Processing Pipeline

The raw detector output consisted of CSV files, each containing time-resolved count data across all energy bins for a single 900-second run. The data processing proceeded through the following sequential steps:

1. Import the CSV data files for all four source trials and the single background trial.
2. Sum counts over the full 900-second integration time for each trial to obtain the total counts per energy bin per trial.
3. Average the four source-trial spectra to reduce statistical fluctuations:
$$N_{\text{avg_source}}(E) = 0.25 * \sum N_i(E).$$
4. Subtract the single background spectrum from the averaged source spectrum to remove environmental radiation contributions: $N_{\text{exp}}(E) = N_{\text{avg_source}}(E) - N_{\text{background}}(E).$
5. Convert the bin index to physical energy (in keV) using a previously determined detector energy calibration (calibrated bin-to-energy conversion factor).

This processing pipeline produced the final background-subtracted experimental spectrum $N_{\text{exp}}(E)$, which became the primary for all subsequent analysis. The total data collection duration was ~900 seconds, and the detector integration time per sample within each trial was 2 seconds.

3. Data and Analysis

3.1 Raw Spectra and Trial Comparison

Figure 1 shows the measured energy spectra for all four source trials (Spectra 0–3) and the single background trial, each plotted as total counts per energy bin over the full 900-second integration. The four source spectra are broadly consistent in shape and magnitude, all exhibiting a sharp rise at low energies (peaking near 150–200 keV) followed by a long exponential-like tail extending to higher energies. The background spectrum is substantially lower in magnitude across all energy bins, confirming that the source is the dominant contributor to the observed counts.

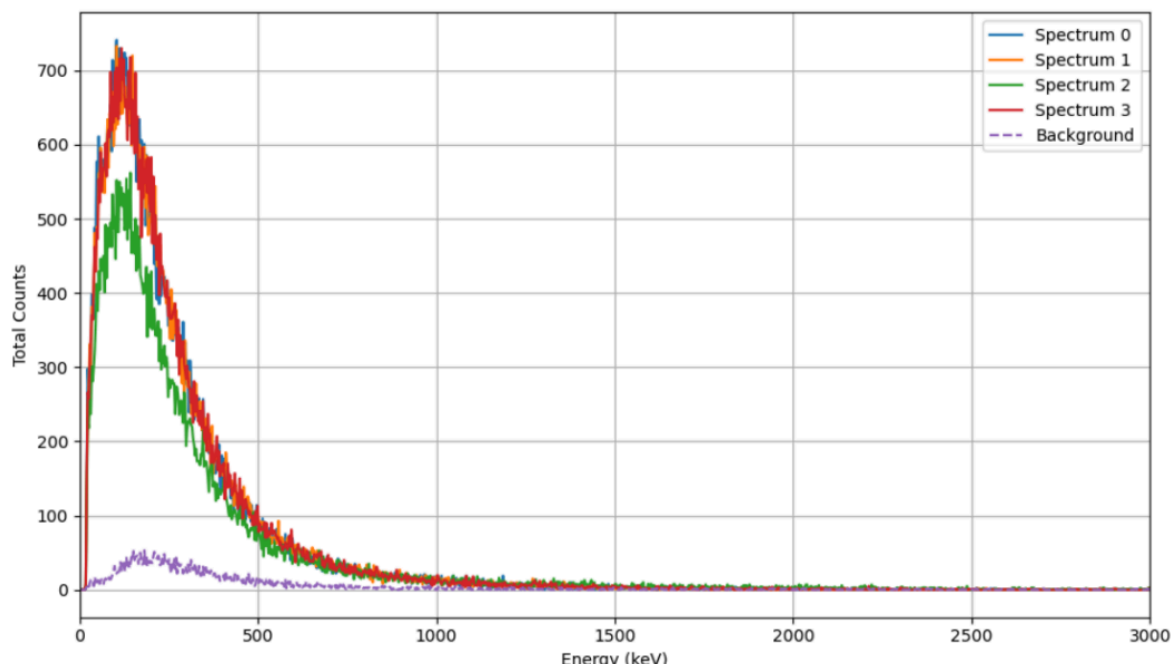


Figure 1: Measured energy spectra for each of the four source trials (Spectra 0–3) and the single background trial. The background spectrum is shown as a purple, dashed line.

Spectrum 2 visibly exhibits a lower count magnitude compared to Spectra 0, 1, and 3. As noted in Procedure, this can be attributed to the detector slipping off the source briefly during the collection period, reducing the effective detection solid angle. Including this trial in the average slightly reduces the average count rate but still improves statistical precision compared to using only three trials, since the underlying spectrum shape of Spectrum 2 remains consistent with the others outside the affected interval and the slip was not for a substantial time nor had an affect the energy deposition angle in the detector, altering the bin energy deposition.

3.2 Comparison of Experimental Spectrum to Theory

Figure 2a shows the background-subtracted, averaged experimental spectrum $N_{\text{exp}}(E)$ overlaid with the normalized combined theoretical Sr-90/Y-90 beta spectrum. The theoretical spectrum has been scaled by a single multiplicative constant so that the total integrated counts within the comparison region match the experimental spectrum, allowing the spectral shape to be directly compared.

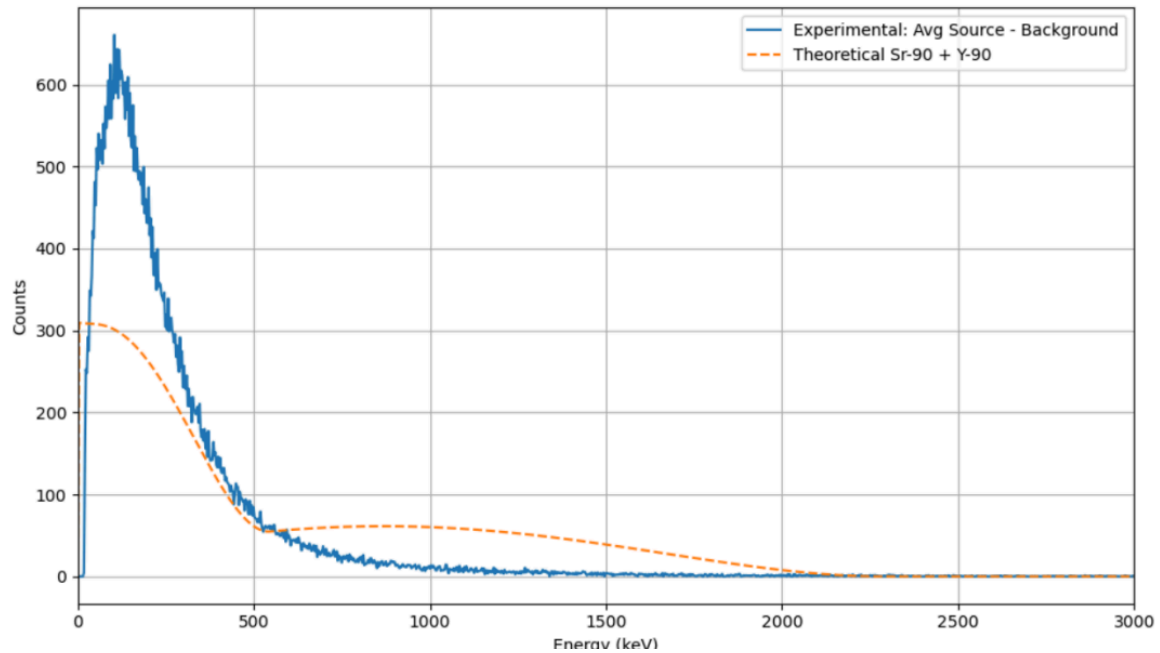


Figure 2a: Background-subtracted average experimental spectrum (blue) compared to the normalized combined theoretical Sr-90/Y-90 beta spectrum (orange dashed). The shape mismatch is what motivated the development of a detector response model.

The discrepancy between theory and experiment is immediately apparent [and is displayed visually in the residuals found in Figure 4 (Results, pg 12)] :

- The theoretical spectrum significantly overestimates counts at higher energies (above roughly 300–500 keV). In the Y-90-dominated energy range above 546 keV, the theoretical prediction falls off gradually while the experimental counts drop steeply.
- The theoretical spectrum underestimates counts at low energies. The experimental spectrum sharply peaked near 150–200 keV, with far more counts in the low-energy bins than the theoretical spectrum predicts.

- The experimental peak is both sharper and located at lower energy than the theoretical peak, suggesting that high-energy electrons are being redistributed downward in energy.

These systematic discrepancies are fully consistent with physical processes occurring within the detector. High-energy electrons entering the crystal are not depositing all of their kinetic energy before exiting the detector; they may escape before fully stopping (electron escape), scatter back out of the detector (backscattering), or radiate a fraction of their energy as bremsstrahlung X-rays that then escape the crystal. In each case, the measured energy is recorded in a lower-energy bin than the true emission energy. The net effect is a downward redistribution of counts from high-energy to low-energy bins, consistent with the observed discrepancy. Again, this motivates the construction of a quantitative detector response model.

3.3 Detector Response Model

An ideal detector would measure the true emitted spectrum $N_{\text{true}}(E)$ directly. A real detector instead measures a convolution of the true spectrum with a detector response function $R(E, E')$ and a detection efficiency $\varepsilon(E')$, according to:

$$N_{\text{det}}(E) = \int \varepsilon(E') \cdot R(E, E') \cdot N_{\text{true}}(E') dE'$$

where E' is the true emitted energy and E is the detected energy. The detector response function $R(E, E')$ describes the probability that a particle emitted with true energy E' is recorded at measured energy E . A semi-empirical detector response function was constructed from three physically motivated sectionaries.

Component 1 — Gaussian Resolution:

When a beta particle deposits energy in the crystal, it produces a finite number of scintillation photons. The exact number fluctuates according to Poisson statistics, and additional electronic noise is introduced in the photodetector readout. As a result, even if all particles deposit exactly the same energy E' , the detector records a distribution of measured energies centered on E' with finite width. This intrinsic detector resolution is modeled as a Gaussian:

$$R_{\text{Gauss}}(E, E') \propto \exp[-(E - E')^2 / (2\sigma^2(E'))]$$

where the energy-dependent resolution width $\sigma^2(E') = \sigma_0^2 + (\sigma_{\text{frac}} \cdot E')^2$ incorporates both a constant baseline term σ_0 – arising from electronic noise and fixed contributions to resolution broadening – and an energy-dependent fractional term σ_{frac} – arising from the statistical nature of scintillation photon production, which tends to worsens at higher energies.

Component 2 — Energy-Loss Tail:

To model incomplete energy deposition (electron escape, backscattering, bremsstrahlung), an asymmetric exponential tail is included for detected energies below the true energy E' . This tail can account for all processes by which a particle deposits less energy than it emitted:

$$R_{tail}(E, E') \propto \exp[-(E' - E) / \lambda(E')] \quad \text{for } E \leq E'$$

where $\lambda(E') = \lambda_0 + \lambda_1 \cdot E'$ is the energy-dependent tail scale. This parameterization allows the tail to grow at higher energies, capturing the physical expectation that more energetic electrons are more likely to travel farther in the crystal and thus more likely to escape or scatter before full deposition.

Component 3 — Detector Efficiency:

A separate efficiency function $\varepsilon(E')$ suppresses detection probability at high energies. This reflects the increasing likelihood that higher-energy electrons exit the detector before depositing all their energy, making full energy detection progressively less probable. The efficiency is modeled as:

$$\varepsilon(E') = 1 / [1 + (E'/E_c)^n]$$

where E_c is the critical energy at which detection efficiency falls to 50%, and n controls the steepness of the efficiency roll-off. This form smoothly transitions from unastic efficiency at low energies to strongly suppressed efficiency at high energies.

The combined detector response function blends the Gaussian and tail components using an energy-dependent mixing fraction $f(E')$:

$$R(E, E') = [1 - f(E')] \cdot R_{Gauss}(E, E') + f(E') \cdot R_{tail}(E, E') \\ \text{where } f(E') = f_0 + f_1 \cdot (E'/E_0)$$

At low energies, $f(E') = f_0 \approx 0$, so the Gaussian response dominates, consistent with electrons being fully absorbed. At high energies, the tail fraction increases, reflecting the growing contribution of incomplete energy deposition.

3.4 Fitting Strategy

The eight free parameters $\{\sigma_0, \sigma_{frac}, \lambda_0, \lambda_1, f_0, f_1, E_c, n\}$ were determined by nonlinear least squares regression using the `scipy.optimize.least_squares` function in Python. The fit was performed over a restricted energy range of $20 \leq E \leq 1200$ keV.

The lower bound of 20 keV was chosen to exclude the detector threshold region, where electronic noise dominates the signal and is not explicitly modeled. The upper bound of 1200 keV was chosen because above this energy, the number of recorded counts becomes very small, leading to large relative statistical uncertainties, where including

these high-uncertainty bins would disproportionately influence the fit in an unphysical way.

To appropriately weight the contribution of each energy bin, Poisson-like residuals were used:

$$r_i = [N_{model}(E_i) - N_{exp}(E_i)] / \sqrt{[N_{exp}(E_i) + 1]}$$

This weighting reflects the idea that Poisson counting uncertainty scales as the square root of the number of counts. Bins with high counts thus have greater influence on the fit, while bins with few counts contribute less. The +1 term in the denominator prevents division by zero in bins with zero measured counts, while still having a minimal effect on the other bins.

The model spectrum was normalized before each fit by scaling the total model counts within the fitting region to match the total experimental counts in that region. This normalization is computed directly from the data rather than treated as a free parameter, ensuring that the optimizer prioritizes shape over amplitude.

All analysis code is available at <https://github.com/gabe6295/E11aga>.

4. Results

4.1 Fitted Model vs. Experimental Spectrum

Figure 3 shows the result of this: the background-subtracted experimental spectrum plotted alongside the fitted detector response model and the unmodified theoretical Sr-90/Y-90 spectrum. The contrast between the theoretical spectrum without detector effects and the fitted model with detector effects makes clear the scale to which the detector influenced the spectrum shape.

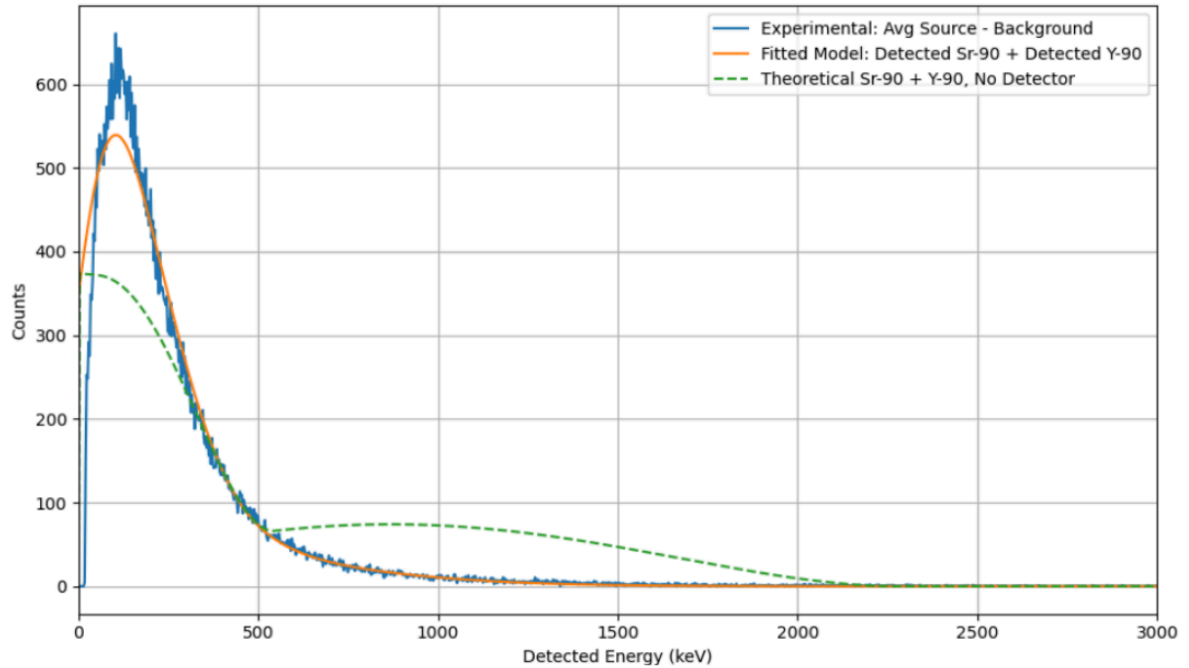


Figure 3: Background-subtracted experimental spectrum (blue) plotted with the fitted detector response model (orange) and the theoretical Sr-90+Y-90 spectrum without detector effects (green dashed)/.

The fitted model achieves strong agreement with the experimental spectrum across the full fitting range. In particular:

- The suppression of high-energy counts (above ~500 keV) is accurately reproduced by the combination of the efficiency rolloff and the energy-loss tail.
- The enhancement of counts at low energies relative to the theoretical spectrum is captured by the redistribution of high-energy events through the tail component of the response function.
- The overall spectral shape, including the sharp peak near 150–200 keV and the rapid fall-off at higher energies, is well matched throughout the fitting region.

Beyond the fitting range ($E > 1200$ keV), the model prediction continues to agree reasonably with the experimental data, though agreement is less critical in this region given the large statistical uncertainties.

4.2 Fitted Parameters and Physical Interpretation

The fitted parameter values from the nonlinear least squares regression are given in Table 1.

Parameter	Fitted Value	Physical Interpretation
σ_0	85.56 keV	Baseline resolution broadening from electronic noise and fixed scintillation fluctuations
σ_{frac}	0.2448	~25% fractional energy-dependent broadening; resolution degrades significantly at higher energies
f_0	6.46×10^{-36}	Negligible constant tail fraction; Gaussian response dominates at low energies
f_1	2	Strong energy-dependent tail; high-energy electrons increasingly likely to escape or scatter
λ_0	1.901 keV	Small constant energy-loss scale; minimal energy loss at threshold
λ_1	0.2001	~20% energy-dependent loss scale; higher-energy electrons lose more energy before detection
E_c	513.6 keV	~500 keV efficiency cutoff; detection probability drops sharply above this energy
n	2.904	Sharp sigmoid-like efficiency roll-off above E_c

Table 1: Best-fit detector response model parameters from nonlinear least squares regression over the energy range 20–1200 keV, with physical interpretation of each parameter.

The fitted parameters are both physically meaningful and generally consistent. The baseline resolution $\sigma_0 = 85.56$ keV is large relative to the energy scale of the spectrum, indicating substantial broadening contributions from electronics noise and intrinsic variability in scintillation light production that are independent of the deposited energy. The fractional resolution $\sigma_{\text{frac}} = 0.2448$ implies that at an energy of 1000 keV, for instance, the Gaussian width from the fractional term alone would be ~245 keV, so resolution indeed degrades strongly at higher energies. This is a known property of CsI scintillators, which have relatively low light yield compared to other detector forms and thus poorer resolution.

The tail fraction parameters $f_0 \approx 0$ and $f_1 = 2$ indicate that the exponential energy-loss tail plays essentially no role at low energies (where electrons fully stop in the crystal) but becomes the dominant component at high energies. Specifically, $f(E') = f_0 + f_1 \cdot (E'/E_0)$ approaches $\sim 2 \cdot (E'/E_0)$ at high energies, meaning the tail fraction is substantial across the Y-90 energy range. Note that at higher energies, $f(E')$ exceeds 1, while in practice, the mixing fraction is clipped to $[0,1]$ in the implementation, so the linear

parameterization serves as an approximation valid over the primary fitting range rather than a physical probability at all energies. The tail scale parameters $\lambda_0 = 1.901$ keV and $\lambda_1 = 0.2001$ indicate that higher-energy electrons lose approximately 20% of their energy per keV of true energy on average before detection, consistent with the physics of electron escape and bremsstrahlung emission in a finite-volume detector.

The efficiency parameters $E_c = 513.6$ keV and $n = 2.904$ indicate that detection probability drops sharply, to near 50% efficiency by ~ 514 keV and continuing to fall steeply above that. This explains why the experimentally measured spectrum is so strongly dominated by low-energy events, where the Y-90 source emits electrons with energies in the 500–2280 keV range, but these are suppressed by both efficiency loss and energy redistribution into lower bins.

4.3 Residuals and Goodness of Fit

Figure 4 shows the residuals between the experimental spectrum and the fitted model, defined as $N_{\text{exp}}(E) - N_{\text{model}}(E)$, plotted as a function of detected energy.

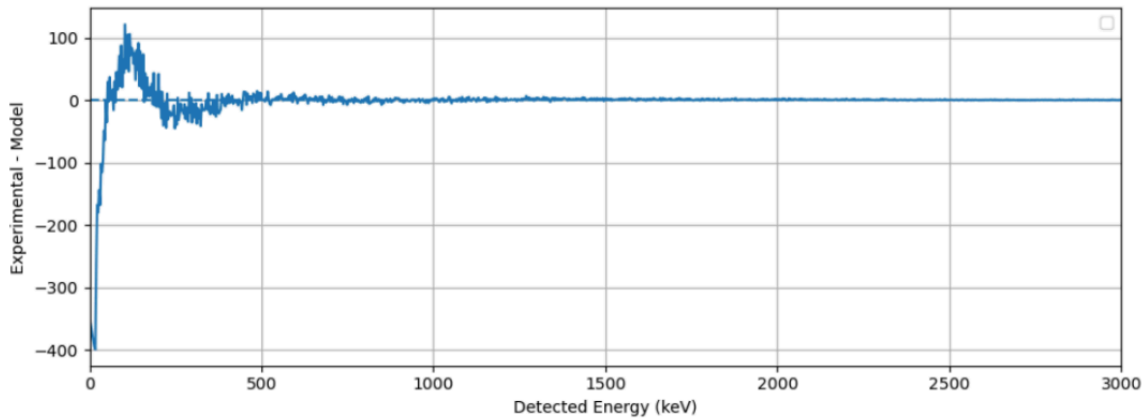


Figure 4: Residuals between the background-subtracted experimental spectrum and the fitted detector response model.

Within the fitting range (20–1200 keV), the residuals are generally scattered about zero, indicating that the model captures the dominant spectral features. Systematic deviations are most prominent at very low energies ($E < 100$ keV), where the model tends to underestimate the measured counts. This suggests the presence of physical effects not fully captured by the current model, including multiple scattering events within the crystal, detector threshold artifacts, or more complex noise contributions that cause a pile-up of events in the lowest energy bins.

The goodness of fit is quantified by the reduced chi-squared statistic, which evaluates the weighted sum of squared residuals normalized by the number of degrees of freedom:

$$\chi^2_{\nu} = 2.49$$

The residuals at very low energies ($E < 100$ keV) are the primary driver of the elevated χ^2_{ν} . To assess whether this region meaningfully affects the physical conclusions, the fitted parameters can be considered in terms of their sensitivity to that energy range. The low-energy tail and Gaussian resolution parameters (λ_0, σ_0) are most constrained by the low-energy bins, while the efficiency and high-energy tail parameters (E_c, n, f_1, λ_1) are determined almost entirely by the 200–1200 keV region where the residuals are small and scatter randomly about zero. The systematic underestimate below 100 keV is therefore unlikely to bias the physically meaningful parameters that govern the conclusions about detector behavior in the Y-90 energy range. The elevated χ^2_{ν} reflects the presence of unmodeled low-energy physics (threshold pile-up, electronic noise contributions) rather than a failure of the model in the regime relevant to the central research question.

A value of $\chi^2_{\nu} = 1$ would indicate a statistically perfect fit under Poisson statistics. The value of 2.49, while above unity, is entirely expected for a semi-empirical model of this complexity. Contributions from multiple scattering, detailed detector geometry, and threshold behavior are not explicitly included. A reduced chi-squared of 2.49 indicates that deviations are on average about 1.58 times the Poisson standard deviation is considered satisfactory given these approximations, and shows that the model captures the dominant physical processes governing the measured spectrum.

5. Discussion

5.1 Detector Response vs. Nuclear Physics

The most striking result of this project is the degree to which the measured beta spectrum differs from the first-principles prediction. Without any detector corrections, the theoretical Sr-90 combined spectrum significantly misrepresents the experimental data, where it predicts a broad peak centered near ~700 keV and substantial counts extending to 2280 keV, while the experiment shows an intense, narrow peak at 150–200 keV and a rapid exponential decrease toward higher energies. The fitted model demonstrates that this apparent contradiction can be entirely explained by detector physics.

This finding has important general implications, showing that the measured spectrum from a finite-volume scintillation detector can be governed almost entirely by detector

response rather than the intrinsic source activity spectrum. An analyst who did not account for these effects could draw entirely incorrect conclusions about the source isotopes present or their relative activities.

5.2 Physical Processes and Their Contributions

The three model components can be interpreted independently to understand their physical contributions to the measured spectrum shape:

Energy resolution broadening (Gaussian component):

The large baseline resolution $\sigma_0 = 85.56$ keV smears the spectrum at the scale of the Sr-90 endpoint energy (546 keV), washing out any distinct spectral feature that might otherwise be visible at the Sr-90 endpoint. The energy-dependent broadening ($\sigma_{\text{frac}} = 0.2448$) blurs further these features at high energies, making spectral structures in the Y-90 range nearly impossible to resolve with this detector.

Incomplete energy deposition (tail component):

The energy-loss tail is the dominant physical effect explaining the strong overrepresentation of low-energy events. As Y-90 emits electrons with energies from 546 keV to 2280 keV, the vast majority of these experience partial energy deposition, as they escape the crystal, backscatter, or radiate via bremsstrahlung, and their energy is recorded in lower-energy bins. The exponential tail parameterization captures this redistribution process and shows that high-energy electrons contribute substantially to the apparent low-energy peak in the experimental spectrum. Without accounting for this, one could incorrectly attribute the large low-energy peak entirely to Sr-90 activity.

Detector efficiency (efficiency component):

The sharp efficiency rolloff above $E_c = 513.6$ keV with exponent $n = 2.904$ further suppresses high-energy event registration. Combined with the tail, this means that Y-90 events above ~500 keV are not only redistributed to lower energies but are also less likely to register a full-energy event even when they do interact with the detector. This is physically consistent with the mean free path of high-energy electrons in CsI, where higher-energy electrons tend to travel farther and are more likely to exit the detector crystal before fully stopping.

5.3 Systematic Uncertainties and Model Limitations

Several sources of systematic uncertainty affect the experimental results:

- Single background trial: Only one background spectrum was collected, compared to four source spectra. This introduces an asymmetric statistical uncertainty into the background subtraction. The background itself has Poisson fluctuations, and averaging over multiple background trials would reduce the associated systematic error. This is the most significant controllable limitation of the current experimental setup.
- Spectrum 2 detector slip: The lower count rate in Spectrum 2 slightly biases the averaged source spectrum downward, introducing a small but systematic underestimate of the true average activity. The effect is expected to be small since the shape of Spectrum 2 is consistent with the others.
- Low-energy model mismatch: The residuals at very low energies ($E < 100$ keV) indicate that the model underestimates the counts in this regime. Physical effects not included in the model, such as detector threshold pile-up, electronic noise events, or multiple scattering contributions that add counts at very low energies, account for this discrepancy.
- Simplified response function: The semi-empirical functional forms chosen for R_{Gauss} , R_{tail} , and $\epsilon(E')$ are physically motivated but are not explicitly derived from first principles. The true response function of the detector has a more complex dependence on detector geometry, crystal dimensions, and the angular distribution of emitted electrons. The current parameterization approximates these effects with a number of free parameters.
- Energy calibration: The conversion from bin index to physical energy relies on a separately calibrated bin-to-energy factor. Any systematic error in this calibration would shift all energy values and distort the comparison between theory and experiment.

5.4 Comparison to Expectations

The project's outcome largely matched initial expectations in qualitative terms, as it was sort of anticipated that detector effects would significantly distort the measured spectrum. The quantitative degree of distortion in the sharpness of the low-energy peak, the complete suppression of counts above ~ 1000 keV, and the large baseline resolution was somewhat more extreme than might be expected from a larger or more efficient detector. This reflects the relatively small size of the CsI crystal used and the high endpoint energy of Y-90, which makes partial energy deposition very prevalent.

The success of the semi-empirical model in fitting the spectrum ($\chi^2_{\nu} = 2.49$, with systematic residuals localized to very low energies) confirms that the three physical

components incorporated in the model are, generally, the dominant processes governing the detector response. No additional components were necessary to achieve a good fit over the primary energy range, though there may be more components that minorly affect the fit.

6. Conclusions and Future Work

6.1 Summary of Findings

This project investigated how the intrinsic beta decay spectrum of a Sr-90 source is modified by the response of a CsI scintillation detector, and whether a parameterized and physically motivated detector response model can quantitatively reproduce the measured spectrum in replacement for a known source subtractant.

The results demonstrate that the experimentally observed beta spectrum differs dramatically from the first-principles theoretical prediction. However, these discrepancies are entirely explained by three physical detector processes:

- Finite energy resolution: intrinsic fluctuations in scintillation photon production and electronic noise broaden each detected energy into a Gaussian distribution.
- Incomplete energy deposition: electron escape, backscattering, and bremsstrahlung radiation redistribute high-energy events into lower-energy bins.
- Detection efficiency reduction: the probability of registering a full-energy event falls sharply above ~514 keV, explaining the severe suppression of high-energy counts even from the high-yield Y-90 component.

By incorporating these three effects into a semi-empirical detector response model and fitting 8 free parameters by nonlinear least squares regression, strong agreement is achieved between the model and experiment over the fitting range 20–1200 keV ($\chi^2_\nu = 2.49$). The fitted parameters are all physically meaningful, consistent with the known properties of CsI scintillators, and mutually reinforcing in their physical interpretation.

The central conclusion is that the observed beta spectrum is governed almost entirely by detector response rather than intrinsic nuclear physics; however, these effects can be adequately accounted for in a free parameters fitted model.

This highlights a fundamental principle of experimental radiation physics, where the measured distribution is a convolution of the true spectrum with the detector response, and correctly interpreting data requires understanding and modeling the detector.

6.2 Assessment of Project Goals

All stated project goals were achieved. Theoretical beta spectra for Sr-90 and Y-90 were constructed using the exact relativistic Fermi function and combined under secular equilibrium. The comparison to experimental data revealed clear, systematic discrepancies attributable to detector physics. A semi-empirical detector response model was developed with physically motivated components and successfully fitted to the data, achieving a satisfactory reduced chi-squared. The fitted parameters yield a coherent physical picture of the CsI detector's energy resolution, energy-loss characteristics, and efficiency.

6.3 Future Work

Several improvements to both the experimental approach and the model would strengthen and extend the results of this project:

- Multiple background trials: The most immediately impactful improvement would be averaging the background over 4 or more trials, as was done for the source spectra. This would reduce the systematic uncertainty in the background subtraction and improve confidence in the residuals at all energies.
- Mechanical detector mount: A fixed mechanical mount for the detector would eliminate the source-detector geometry variations seen in Spectrum 2 and ensure reproducibility between trials.
- Independent energy calibration: Using mono-energetic calibration sources would allow the energy axis to be independently calibrated and the baseline resolution σ_0 to be directly measured rather than fitted.
- Extended fitting range: With better statistics at high energies, the fitting range could be extended above 1200 keV to better constrain the parameters f_1 and E_c , which primarily control the high-energy tail behavior.
- A full particle transport simulation using a GEANT4 frame would model the detector response from first principles, accounting explicitly for detector geometry, crystal dimensions, material properties, and angular dependence of electron scattering. Comparing the GEANT4-predicted response to the semi-empirical model would validate the physical interpretation and potentially reveal additional processes not captured by the current parameterization. This would be in the scope of a more complex upper division course.
- Angular-dependent scattering model: The current model does not account for the angular distribution of electrons entering the detector or the angular dependence of escape probability. A more realistic model would incorporate these geometric factors.

Overall, this project demonstrates that a physically motivated, semi-empirical detector response model can successfully bridge between theoretical beta decay spectra and experimentally measured distributions. The work reflects a methodology of experimental nuclear physics, in constructing and fitting a forward model that propagates from the underlying physics through all instrumental effects to the observed data, and allowing meaningful physical conclusions to be drawn even when the raw data appears far removed from the underlying theory.

7. Works Cited

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